Uncertainty-Gated Dual-Stage PEFT: Trustworthy Few-Shot Plant Disease Recognition on Edge Devices

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***Abstract*—Automated plant disease diagnosis via deep learning is critical for precision agriculture and early crop protection. However, existing vision models heavily rely on large-scale annotated datasets and struggle with low-data regimes (few-shot settings) when exposed to novel plant pathogens. In this work, we propose an Uncertainty-Gated Dual-Stage Parameter-Efficient Fine-Tuning (PEFT) framework based on Vision-Language Pre-training (CLIP). In Stage 1, we introduce low-rank adapters (LoRA) into the vision backbone to capture domain-specific botanical morphology while preventing catastrophic forgetting. In Stage 2, for novel pathogens with only $K = 5$ support samples, we synthesize knowledge-driven text prototypes alongside empirical visual prototypes and design an entropy-based predictive uncertainty gate. Confident predictions are directly routed via prototype similarity, whereas ambiguous queries pass through a non-linear residual feature adapter trained with entropy sharpening. Empirical evaluations on the PlantVillage dataset demonstrate substantial gains over zero-shot baselines and conventional adapter architectures, while dynamic INT8 quantization confirms real-time feasibility for edge deployment.**

***Index Terms*—Few-Shot Learning, Parameter-Efficient Fine-Tuning (PEFT), Vision-Language Models, CLIP, Plant Pathology, Uncertainty Gating, Edge AI.**

I. Introduction

Food insecurity remains a major global challenge, with more than 700 million people experiencing hunger worldwide. Plant diseases are among the factors that threaten agricultural productivity and food security, contributing to substantial crop losses and economic damage. Estimates indicate that plant pests and diseases are responsible for annual losses of up to 40% of global crop production. Conventional plant disease diagnosis, which often relies on manual visual inspection, can be challenging in agricultural environments because different diseases may exhibit similar or overlapping visual symptoms. Such similarities can lead to misdiagnosis and delayed intervention, potentially worsening disease progression and reducing crop yields.

To address these challenges, computer vision and artificial intelligence (AI) have become important tools for automated plant disease detection in smart agriculture. However, traditional deep learning methods usually require large amounts of labeled training data to achieve good performance. In real-world agricultural settings, collecting such data can be difficult, particularly for rare or newly emerging diseases for which only a limited number of images are available. Moreover, collecting and manually labeling thousands of images for each disease is time-consuming and requires considerable effort, making it difficult to apply these methods to new diseases.

Few-shot learning (FSL) offers a potential solution to this problem by allowing models to recognize new plant disease categories using only a small number of labeled examples. Recent studies have explored different approaches to this task, including metric-based methods, meta-learning techniques, and multimodal models. Although these methods have shown promising results on benchmark datasets, their performance may vary when applied to images collected under different field conditions. In addition, some approaches require complex architectures or substantial computational resources, which can make practical deployment more challenging.

To overcome these challenges, Vision-Language Models (VLMs) such as CLIP [1] offer strong transfer capabilities. However, off-the-shelf VLMs often struggle on fine-grained lesion differentiations in low-data regimes. In this paper, we propose an **Uncertainty-Gated Dual-Stage Parameter-Efficient Fine-Tuning (PEFT)** framework tailored for low-resource plant pathology. Our contributions are threefold:

- **Dual-Stage PEFT Architecture:** We combine parameter-efficient LoRA tuning [2] on base agricultural categories with a residual few-shot adapter for novel categories.
- **Entropy-Driven Uncertainty Gating:** We formulate an adaptive routing mechanism based on Shannon entropy that dynamically decides whether a query image requires deep feature adaptation or direct prototype matching.
- **Edge Quantization Benchmarking:** We evaluate post-training dynamic INT8 quantization, demonstrating that the full inference pipeline achieves real-time execution on standard edge CPUs.

II. Methodology

A. Stage 1: Contrastive Base-Domain Adaptation

Given a pre-trained CLIP model with vision encoder $f_v(\cdot)$ and text encoder $f_t(\cdot)$, we inject low-rank decomposition

matrices $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{d \times r}$ into the multi-head self-attention projections with rank $r \ll d$. The model is optimized on base categories $\mathcal{Y}_{\text{base}}$ using symmetric InfoNCE contrastive loss:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{2B} \sum_{i=1}^{B} \left[\log \frac{e^{\tau \langle v_i, t_i \rangle}}{\sum_j e^{\tau \langle v_i, t_j \rangle}} + \log \frac{e^{\tau \langle v_i, t_i \rangle}}{\sum_j e^{\tau \langle v_j, t_i \rangle}} \right] \tag{1}$$

where τ is a learnable temperature scale parameter and $\langle \cdot, \cdot \rangle$ denotes cosine similarity.

B. Multimodal Prototype Construction

For novel classes $\mathcal{Y}_{\text{novel}}$, we extract multimodal representations:

- **Knowledge-Driven Text Prototypes (P_t):** Synthesized by averaging embeddings from engineered diagnostic prompts describing specific symptomatic lesions (e.g., margins, pustules, halos).
- **Visual Support Prototypes (P_v):** Formed by computing normalized centroid embeddings from the K-shot support set $\mathcal{S}$:

$$P_v^{(c)} = \text{Normalize}\left(\frac{1}{K} \sum_{x \in \mathcal{S}_c} f_v(x) \right) \tag{2}$$

C. Predictive Uncertainty Gating

For an input image x, direct logits are computed against vision prototypes: $z_{\text{direct}} = \tau \cdot f_v(x) P_v^T$. The softmax probability vector $p = \text{softmax}(z_{\text{direct}})$ yields normalized predictive entropy:

$$\mathcal{H}(p) = -\frac{1}{\log |\mathcal{Y}_{\text{novel}}|} \sum_{c=1}^{|\mathcal{Y}_{\text{novel}}|} p_c \log(p_c + \epsilon) \tag{3}$$

The confidence score is defined as $C(x) = 1.0 - \mathcal{H}(p)$. An adaptive gate mask is triggered:

$$g(x) = \mathbb{I}\left(C(x) \geq \theta\right) \tag{4}$$

where $\theta \in [0, 1]$ is a learnable uncertainty threshold. Highly confident queries ($g(x) = 1$) bypass adapter overhead, while uncertain queries ($g(x) = 0$) are routed through a non-linear residual bottleneck adapter.

III. Experimental Results

TABLE I
Ablation Study on 5-Shot Novel Plant Disease Recognition (PlantVillage)

Method / Configuration	Accuracy (%)	Macro F1 (%)
1. Zero-Shot Baseline CLIP	68.42	65.18
2. Dual-Stage PEFT (w/o Gate)	84.15	82.70
3. Proposed Uncertainty-Gated PEFT	**89.60**	**88.45**

As summarized in Table I and Table II, the proposed uncertainty-gated mechanism delivers a superior trade-off between predictive accuracy and computational latency, retaining over 99.5% of the original accuracy after dynamic INT8 quantization.

TABLE II
Edge AI Deployment & Quantization Benchmarking (CPU)

Model Variant	Size (MB)	Latency (ms)	Acc. (%)
FP32 Baseline (Merged LoRA)	350.2	48.6	89.60
INT8 Quantized Edge Model	**88.5**	**19.2**	**89.15**

IV. Conclusion

We presented an uncertainty-gated dual-stage PEFT framework for few-shot plant disease classification. By combining contrastive base tuning with entropy-guided multimodal prototype routing, our method achieves state-of-the-art few-shot accuracy while enabling low-latency deployment on edge agricultural devices.

References

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